Paper D-13: Population Genetics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ, constraint C, surface responsiveness S, and structural memory M_s. The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Evolution is the algorithm by which life explores morphological and functional space. Mutations introduce random variations, gene flow mixes them across geography, and natural selection eliminates variants that are ill-suited to the environment.

In CDFD, this process is entirely isomorphic to fluid dynamics. The genome of a population is a fluid network of information passing through time. The environment acts as a restrictive channel that shapes and squeezes this flow.

3 The Gene Pool as an Adaptive Surface

We track the evolutionary stability of a species via the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

In the context of population genetics:

- **Flux (Φ)**: The rate of genetic variation introduced by mutation, recombination, and immigration (gene flow). This is the raw informational throughput of the species.
- **Constraint (C)**: Natural selection and environmental pressure (predators, climate, resource scarcity). Selection acts as a filter, removing non-viable informational flow.
- **Responsiveness (S)**: The genetic diversity and phenotypic plasticity of the population. A highly diverse population can easily flex to meet changing constraints.
- **Memory (M_s)**: Genetic fixation and canalization. As a trait becomes heavily selected for, it becomes fixed in the genome. The species "remembers" successful past adaptations, severely restricting future morphological deviations (high M_s).

3.1 Extinction vs. Speciation

A well-adapted species in a stable environment operates at $\Psi_s \approx 1$. The genetic flux (Φ) perfectly matches the environmental constraints (C), and the genetic memory (M_s) maintains the optimal phenotype.

If the environment suddenly changes (e.g., rapid climate change), the constraint C spikes violently. If the species has low diversity (S is low) and its genome is heavily canalized (M_s is locked), it cannot adapt. The ratio plummets to $\Psi_s \ll 1$. The flow of genes through time ceases entirely—this is **Extinction**.

Conversely, consider a population spread across a vast area that is suddenly divided by a mountain range. The gene flow (Φ) between the two groups drops to zero. Isolated, each group faces unique local constraints (C). To restore $\Psi_s \approx 1$ locally, the responsiveness (S) of each group explores different mutational pathways. Over time, their genetic memories (M_s) lock into incompatible configurations. The singular adaptive surface has bifurcated into two distinct species. This is **Speciation**, a direct topological consequence of severing the flux network.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Conclusion

The mathematics of evolution are the mathematics of constrained flow. By applying CDFD to population genetics, we eliminate the false dichotomy between biology and physics. The gene pool flows, adapts, and breaks around environmental constraints in the exact same manner that a river carves a canyon or a language splinters into dialects.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

6 Reference Layer

References

[1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.

[2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.

[3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.

[4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.